Replicating Table 1, Panel B from Chernozhukov et al. (2017)

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1 The Original Paper

Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, and Newey (2017), "Double/Debiased/Neyman Machine Learning of Treatment Effects," *American Economic Review: Papers & Proceedings*, 107(5): 261–265, introduces the Double/Debiased Machine Learning (Double ML) estimator for causal inference in settings where high-dimensional nuisance parameters must be estimated. The central research question is how to obtain valid, root-N consistent estimates of a treatment effect parameter θ_0 when the confounding functions are estimated using flexible machine learning methods that introduce regularization bias.

The empirical strategy rests on two ingredients. First, the estimating equation is made Neyman-orthogonal, meaning it is locally insensitive to small errors in the nuisance parameter estimates. Second, cross-fitting is used to split the sample into folds, training the nuisance functions on held-out data to avoid overfitting bias. Together these two steps remove the regularization bias that would otherwise contaminate a naive plug-in estimator.

2 The Result of Interest

The specific result I replicate is the Mean Average Treatment Effect (ATE) reported in Table 1, Panel B, row "ATE (2 fold)," Ensemble column of the online appendix. This result corresponds to the partially linear model with 2-fold cross-fitting. The paper reports a Mean ATE of $9,043 with a standard error of 1,432.

This result is the headline finding of the empirical illustration: after flexibly controlling for income and other confounders using an ensemble of machine learning methods, 401(k) eligibility raises net financial assets by roughly $9,000. It is the cleanest single number to target because it corresponds to the default specification in the replication code (`Master.R`), uses the partially linear model, and focuses on the Ensemble method, which combines the machine learning approaches.

3 Data and Methods

The data come from the 1991 Survey of Income and Program Participation (SIPP), the same dataset used in Chernozhukov and Hansen (2004). The sample contains 9,915 observations. The outcome variable `net_tfa` is net total financial assets, measured in dollars. The treatment variable `e401` is a binary indicator for eligibility to enroll in a 401(k) plan. The nine covariates used in the model are age, income, years of education, family size, and binary indicators for married status, two-earner household, defined benefit pension, IRA participation, and home ownership.

`preprocess.R` loads the raw Stata file `sipp1991.dta` from `input/`, selects the eleven relevant variables, and writes a clean CSV to `temp/clean_data.csv`. `analysis.R` loads that file, sources the original helper functions `ML_Functions.R` and `Moment_Functions.R` from `input/`, and calls `DoubleML()` with the partially linear model specification and 2-fold cross-fitting. To reduce computation time, the estimation is repeated over 10 random sample splits rather than the 100 splits used in the original paper. Parallel computation across 4 cores is used to accelerate estimation. The Mean ATE, Median ATE, and their standard errors are computed following the paper's formulas exactly. Results are written to `output/tables/main_result.tex`, which is `\input`ed directly into this document.

4 Replication Results

Table 1: Replication of Table 1, Panel B (Partially Linear Model, 2-fold) from Chernozhukov et al. (2017)

	This Replication	Paper (Ensemble)
Mean ATE ($)	8970.03	9,043
Std. Error	1406.97	1,432
95% CI	[6212.38, 11727.68]	—
Median ATE ($)	9017.29	9,061
Median SE	1406.97	1,343

Note: Based on 10 sample splits, 2-fold cross-fitting.

Paper values from Table 1, Panel B, Ensemble column.

The replication estimate reported in Table 1 closely matches the corresponding result in Chernozhukov et al. (2017). The estimated average treatment effect of 401(k) eligibility on net financial assets is very similar to the published estimate, and the associated standard errors are nearly identical. Given that the original study averages over 100 random sample splits while this replication uses only 10, small differences of this magnitude are expected. Overall, the replication successfully reproduces the paper's central empirical finding that 401(k) eligibility substantially increases net financial assets.

5 What I Learned

Once the helper functions were sourced correctly, reproducing the estimation was straightforward. The hardest part was path management: the original code assumed all files lived in the same working directory, while the Gentzkow-Shapiro structure separates `input/`, `code/`, and `temp/`. In particular, `Moment_Functions.R` internally sources `ML_Functions.R` with a bare filename, which required patching to use an absolute path. This is a common reproducibility issue: code that works in the author's environment can break when placed in a different directory structure.

More broadly, this replication taught me that Double ML is remarkably stable: six very different machine learning methods (RLasso, Trees, Random Forest, Boosting, Neural Net, Ensemble) all produce estimates in the $7,900 - -$9,200 range, with substantial overlap in their confidence intervals. This stability is exactly what the Neyman orthogonality theory predicts, and seeing it empirically is more convincing than any theorem. If I were the original authors, the one thing that would have made replication easier is absolute rather than relative paths in the helper files, along with a brief README note specifying how the scripts should be run.

References

[1] Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., and Newey, W. (2017). "Double/Debiased/Neyman Machine Learning of Treatment Effects." *American Economic Review: Papers & Proceedings*, 107(5): 261–265.